

1 Module calendar: important dates and overview

The module calendar provides detailed information to students on how to prepare for each contact session. Preparing before class is key for actively participating in class discussions and activities. The goal is to have everyone ready to engage when class starts.

EBN111 LECTURE SCHEDULE 2025

XY – Prof. Xianming Ye; ST – Sakhile Twala; AR – Archie Rohde; AL – Assistant Lecturers
Venues: **Mon** - AE du Toit Auditorium; **Tue** - AE du Toit Auditorium; **Thu**: Roos hall; **Fri**: AE du Toit Auditorium

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
1	Monday 08:30 – 09:20	10/02/2025	AE du Toit Auditorium	ST	Lecture 1	Introduction and Study Guide Study Ch 1.1-1.4 and do textbook Examples: 1.1, 1.2, 1.3 Units, Charge, Current, Voltage Do textbook Practice Problems: 1.1, 1.2, 1.3 Do AMS Assignment #1 (Deadline: 17 February 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	11/02/2025	AE du Toit Auditorium	ST	Lecture 2	Study Ch 1.5-1.6 and do textbook Examples: 1.4, 1.5, 1.6, 1.7 Power, Energy and Circuit Elements Do textbook Practice Problems: 1.4, 1.5, 1.6, 1.7
	Thursday 11:30 – 12:20	13/02/2025	Roos hall	ST	Lecture 3	Study Ch 2.2-2.3 and do textbook Examples: 2.1, 2.2, 2.3, 2.4 Ohm's Law, Nodes & Branches & Loops Do textbook Practice Problems: 2.1, 2.2, 2.3, 2.4
	Friday 07:30 – 08:20	14/02/2025	AE du Toit Auditorium	ST & AL	Tut class #1	Pre-practical Lecture 1 & AMS tutorial Do chapter problems: 1.4, 1.6, 1.8, 1.11, 1.12, 1.16, 1.18, 1.20, 1.24, 1.28, 2.6, 2.7 Do Pre-practical 1 Assignment and submit on AMS (Deadline: 21 February 2025 at 11:59pm)

2	Monday 08:30 – 09:20	17/02/2025	AE du Toit Auditorium	ST	Lecture 4	Study Ch 2.4 and do textbook Examples: 2.5, 2.6, 2.7, 2.8 Kirchhoff's Law Do textbook Practice Problems: 2.5, 2.6, 2.7, 2.8 Do AMS Assignment #2 (Deadline: 24 February 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	18/02/2025	AE du Toit Auditorium	ST	Lecture 5	Study Ch 2.5-2.6 and do textbook Examples: 2.9, 2.10, 2.11, 2.12, 2.13 Series and Voltage division, Parallel and Current division Do textbook Practice Problems: 2.9, 2.10, 2.11, 2.12, 2.13
	Thursday 11:30 – 12:20	20/02/2025	Roos hall	ST	Lecture 6	Study Ch 2.7 and do textbook Examples: 2.14, 2.15 Wye-Delta transformations Do textbook Practice Problems: 2.14, 2.15
	Friday 07:30 – 08:20	21/02/2025	AE du Toit Auditorium	ST, AR & AL	Tut class #2	Collect component packs Do chapter problems: 2.15, 2.22, 2.23, 2.25, 2.32, 2.45b, 2.51b, 2.53b

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
3 Practical 1 ST	Monday 08:30 – 09:20	24/02/2025	AE du Toit Auditorium	ST	Lecture 7	Study Ch 3.2 and do textbook Examples: 3.1, 3.2 Nodal Analysis Do textbook Practice Problems: 3.1, 3.2 Do AMS Assignment # 3 (Deadline: 03 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	25/02/2025	AE du Toit Auditorium	ST	Lecture 8	Study Ch 3.3 and do textbook Examples: 3.3, 3.4 Nodal Analysis with Voltage Sources Do textbook Practice Problems: 3.3, 3.4
	Thursday 11:30 – 12:20	27/02/2025	Roos hall	ST	Lecture 9	Study Ch 3.4 and do textbook Examples: 3.5, 3.6 Mesh Analysis Do textbook Practice Problems: 3.5, 3.6
	Friday 07:30 – 08:20	28/02/2025	AE du Toit Auditorium	AL	Tut class #3	Do chapter problems: 3.2, 3.4, 3.12, 3.16, 3.20, 3.30, 3.31
4	Monday 08:30 – 09:20	03/03/2025	AE du Toit Auditorium	AR	Lecture 10	Study Ch 3.5 and do textbook Examples: 3.7 Mesh Analysis with Current Sources Do textbook Practice Problems: 3.7 Do AMS Assignment # 4. (Deadline: 10 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	04/03/2025	AE du Toit Auditorium	AR	Lecture 11	Study Ch 3.6-3.7 and do textbook Examples: 3.8, 3.9 Nodal and Mesh Analysis by Inspection Do textbook Practice Problems: 3.8, 3.9
	Thursday 11:30 – 12:20	06/03/2025	Roos hall	AR	Lecture 12	Study Ch 3.9 and do textbook Examples: 3.12, 3.13 Transistors Do textbook Practice Problems: 3.12, 3.13
	Friday 07:30 – 08:20	07/03/2025		AR & AL	Class Test 1	Chapters 1, 2 & 3.2-3.3

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
5	Monday 08:30 – 09:20	10/03/2025	AE du Toit Auditorium	XY	Lecture 13	Study Ch 4.2-4.3 and do textbook Examples: 4.1, 4.2, 4.3, 4.4, 4.5 <i>Linearity and Superposition</i> Do textbook Practice Problems: 4.1, 4.2, 4.3, 4.4, 4.5 Do AMS Assignment # 5 (Deadline: 17 March 2025 at 11:59pm).
	Tuesday 13:30 – 14:20	11/03/2025	AE du Toit Auditorium	XY	Lecture 14	Study Ch 4.4 and do textbook Examples: 4.6, 4.7 <i>Source Transformation</i> Do textbook Practice Problems: 4.6, 4.7
	Thursday 11:30 – 12:20	13/03/2025	Roos hall	XY	Lecture 15	Study Ch 4.5 and do textbook Examples: 4.8, 4.9, 4.10 <i>Thevenin's Theorem</i> Do textbook Practice Problems: 4.8, 4.9, 4.10
	Friday 07:30 – 08:20	14/03/2025	AE du Toit Auditorium	AL	Tut class #4	Pre-practical Lecture 2 Do chapter problems: 3.37, 3.44, 3.60, 3.69, 3.72, 3.90, 3.91 Do Pre-practical 2 Assignment and submit on AMS (Deadline: 20 March 2025 at 11:59pm).
6	Friday Timetable 07:30 – 08:20	17/03/2025	AE du Toit Auditorium	AL	Tut class #5	Do chapter problems: 4.4, 4.5, 4.11, 4.18, 4.24, 4.37, 4.38, 4.40 Do AMS Assignment # 5 (Deadline: 31 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	18/03/2025	AE du Toit Auditorium	XY	Lecture 16	Study Ch 4.6 and do textbook Examples: 4.11, 4.12 <i>Norton's Theorem</i>

						Do textbook Practice Problems: 4.11, 4.12
	Thursday 11:30 – 12:20	20/03/2025	Roos hall	XY	Lecture 17	Study Ch 4.8 and do textbook Examples: 4.13 Maximum Power Transfer Do textbook Practice Problems: 4.13
	Friday	21/03/2025	Public Holiday			
7	Thursday 13:30 – 15:00	27/03/2025	Engineering 3-1 Engineering 3-2 Engineering 3-6 Engineering 3-7	Invigilators	Semester Test 1	Test covers Chapters 1-3. ClickUP announcements will be posted with further details.

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
8 Practical 2 AR	Monday 08:30 – 09:20	31/03/2025	AE du Toit Auditorium	AR	Lecture 18	Study Ch 5.2-5.3 and do textbook Examples: 5.1, 5.2 Op Amps, Ideal Op Amps Do textbook Practice Problems: 5.1, 5.2 Do AMS Assignment # 6 (Deadline: 07 April 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	01/04/2025	AE du Toit Auditorium	AR	Lecture 19	Study Ch 5.4-5.5 and do textbook Examples: 5.3, 5.4, 5.5 Inverting and Non-inverting Amplifiers Do textbook Practice Problems: 5.3, 5.4, 5.5
	Thursday 11:30 – 12:20	03/04/2025	Roos hall	AR	Lecture 20	Study Ch 5.6-5.7 and do textbook Examples: 5.6, 5.7, 5.8 Summing and Difference Amplifiers Do textbook Practice Problems: 5.6, 5.7, 5.8
	Friday	04/04/2025	AE du Toit	AL	Tut class #6	Do chapter problems 4.48, 4.52, 4.71, 4.72, 5.6, 5.8, 5.9, 5.13, 5.20, 5.31, 5.32, 5.39, 5.45

	07:30 – 08:20		Auditorium			
9	Monday 08:30 – 09:20	07/04/2025	AE du Toit Auditorium	AR	Lecture 21	Study Ch 5.8 and do textbook Examples: 5.9, 5.10 Cascade Circuits Do textbook Practice Problems: 5.9, 5.10 Do AMS Assignment # 7 (Deadline: 22 April 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	08/04/2025	AE du Toit Auditorium	AR	Lecture 22	Study Ch 6.2-6.3 and do textbook Examples: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7 Capacitors, Series and Parallel Capacitors Do textbook Practice Problems: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7
	Thursday 11:30 – 12:20	10/04/2025	Roos hall	AR	Lecture 23	Study Ch 6.4-6.5 and do textbook Examples: 6.8, 6.9, 6.10, 6.11, 6.12 Inductors, Series and Parallel Inductors Do textbook Practice Problems: 6.8, 6.9, 6.10, 6.11, 6.12
	Friday 07:30 – 08:20	11/04/2025	AE du Toit Auditorium	AL	Tut class #7	Do chapter problems: 5.57, 5.70, 6.10, 6.12, 6.18, 6.40, 6.46, 6.54
	14/04/2025 – 18/04/2025 UP Recess					

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
10	Monday	21/04/2025				Public Holiday
	Monday Timetable 08:30 – 09:20	22/04/2025	AE du Toit Auditorium	ST	Lecture 24	Study Appendix B and do textbook Examples: B.1, B.2, B.3, B.4, B.5 Complex numbers Do textbook Practice Problems: B.1, B.2, B.3, B.4, B.5 Do AMS Assignment # 8 (Deadline: 29 April 2025 at 11:59pm).
	Thursday 11:30 – 12:20	24/05/2025	Roos hall	ST	Lecture 25	Study Ch 9.2-9.3 and do textbook Examples: 9.1, 9.2 Sinusoids and Phasors Do textbook Practice Problems: 9.1, 9.2
	Friday 07:30 – 08:20	25/05/2025		ST & XY	Class Test 2	Chapters 4, 5, 6 Do Pre-practical 3 Assignment and submit on AMS (Deadline: 02 May 2025 at 11:59pm).
11	Monday	28/04/2025				Public Holiday
	Tuesday 13:30 – 14:20	29/04/2025	AE du Toit Auditorium	XY	Lecture 26	Study Ch 9.3-9.4 and do textbook Examples: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8 Phasors and Phasor Relationships with Circuit Elements Do textbook Practice Problems: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8
	Monday Timetable 08:30 – 09:20	30/04/2025	AE du Toit Auditorium	XY	Lecture 27	Study Ch 9.5-9.7 and do textbook Examples: 9.9, 9.10, 9.11, 9.12 Impedance, Kirchhoff's Laws in Frequency Domain Do textbook Practice Problems: 9.9, 9.10, 9.11, 9.12
	Thursday	01/05/2025				Public Holiday
	Friday 07:30 – 08:20	02/05/2025	AE du Toit Auditorium	ST & AL	Tut class # 8	Pre-Practical Lecture 3 Do chapter problems: 9.2, 9.6, 9.13, 9.19, 9.20, 9.42, 9.48, 9.54, 9.66, 9.70

12	Wednesday 15:30 – 17:00	07/05/2025	Engineering 3-1 Engineering 3-2 Engineering 3-6 Engineering 3-7	Invigilators	Semester Test 2	Test covers Chapters 4, 5, 6, 9. ClickUP Announcements will be posted with further details.
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Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
13 Practical 3 ST	Monday 08:30 – 09:20	12/05/2025	AE du Toit Auditorium	XY	Lecture 28	Study Ch 10.2 and do textbook Examples: 10.1, 10.2 Nodal Analysis Do textbook Practice Problems: 10.1, 10.2 Do AMS Assignment # 9 (Deadline: 19 May 2025 at 11:59pm).
	Tuesday 13:30 – 14:20	13/05/2025	AE du Toit Auditorium	XY	Lecture 29	Study Ch 10.3-10.4 and do textbook Examples: 10.3, 10.4, 10.5, 10.6 Mesh Analysis, Superposition Do textbook Practice Problems: 10.3, 10.4, 10.5, 10.6
	Thursday 11:30 – 12:20	15/05/2025	Roos hall	XY	Lecture 30	Study Ch 10.5-10.6 and do textbook Examples: 10.7, 10.8, 10.9, 10.10 Source Transformation, Thevenin & Norton Do textbook Practice Problems: 10.7, 10.8, 10.9, 10.10
	Friday 07:30 – 08:20	16/05/2025	AE du Toit Auditorium	AL	Tut class # 9	Do chapter problems: 10.15, 10.19, 10.36, 10.46, 10.52, 10.56, 10.61, 10.66, 10.68
14	Monday	19/05/2025	AE du Toit Auditorium	ST	Semester Test 3	Optional Semester Test Test covers chapters 6, 9 and 10. ClickUP Announcements will be posted with further details.

	Wednesday	21/05/2025		ST & AL	Aegrotat/Supplementary Pre-practical	Do Sick/Supp Pre-practical Assignment and submit on AMS (Deadline: 26 May 2025 at 11:59pm). ClickUP Announcements will be posted with further details.
15	Monday - Thursday	26/05/2025 – 29/05/2025	CAEC Labs	ST & AL	Aegrotat/Supplementary Lab Practicals.	ClickUP Announcements will be posted with further details.
31/05/2025 – 19/06/2025 Exam Season						